

Fine-tuning LLMs on Benchmark Datasets:

A Data Efficiency Study using Gemma 3 1B and Phi-4 Mini

Datasets: MMLU (Knowledge) | HumanEval (Coding) | GPU: Kaggle P100

1. Overview

This investigates how much training data is actually needed to improve a language model's performance on standard benchmarks. Instead of using massive datasets, we fine-tune two modern models with small amounts of data (50 to 1000 samples) to see how efficiently they can learn.

The core question we answer: Does fine-tuning with limited data help or hurt model accuracy?

Models Tested:

- Gemma 3 1B — Google's small 1 billion parameter model
- Phi-4 Mini — Microsoft's efficient 3.8 billion parameter model

Datasets Used:

- MMLU (Massive Multitask Language Understanding) — Tests general knowledge across 57 subjects like history, science, math, and law
- HumanEval — Tests coding ability with 164 Python programming problems

Why This Approach?

Full fine-tuning of large models requires hundreds of thousands of samples and expensive GPUs. We used QLoRA (Quantized Low-Rank Adaptation) — a technique that reduces memory usage by 75% while still updating the model. This allowed us to run all experiments on a single Kaggle P100 GPU (16GB) which is freely available to anyone.

2. Experimental Setup

Setting	Gemma 3 1B	Phi-4 Mini
Dataset	MMLU + HumanEval	MMLU
Sample Sizes	50, 200, 500	200, 500, 1000
Fine-tuning Method	QLoRA (4-bit)	QLoRA (4-bit)
LoRA Rank	r=16	r=8
Batch Size	2	1
Epochs	3	3
GPU	Kaggle P100 16GB	Kaggle P100 16GB

3. Results

3.1 Gemma 3 1B on MMLU

Experiment	Samples	Accuracy (%)
Baseline (no fine-tuning)	0	30.40%
Fine-tuned	50	29.00%
Fine-tuned	200	29.20%
Fine-tuned	500	28.20%

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FINAL RESULTS SUMMARY
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Experiment                Accuracy
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Baseline (0 samples)      30.40%
Fine-tuned (50 samples)   29.00%
Fine-tuned (200 samples)  29.20%
Fine-tuned (500 samples)  28.20%
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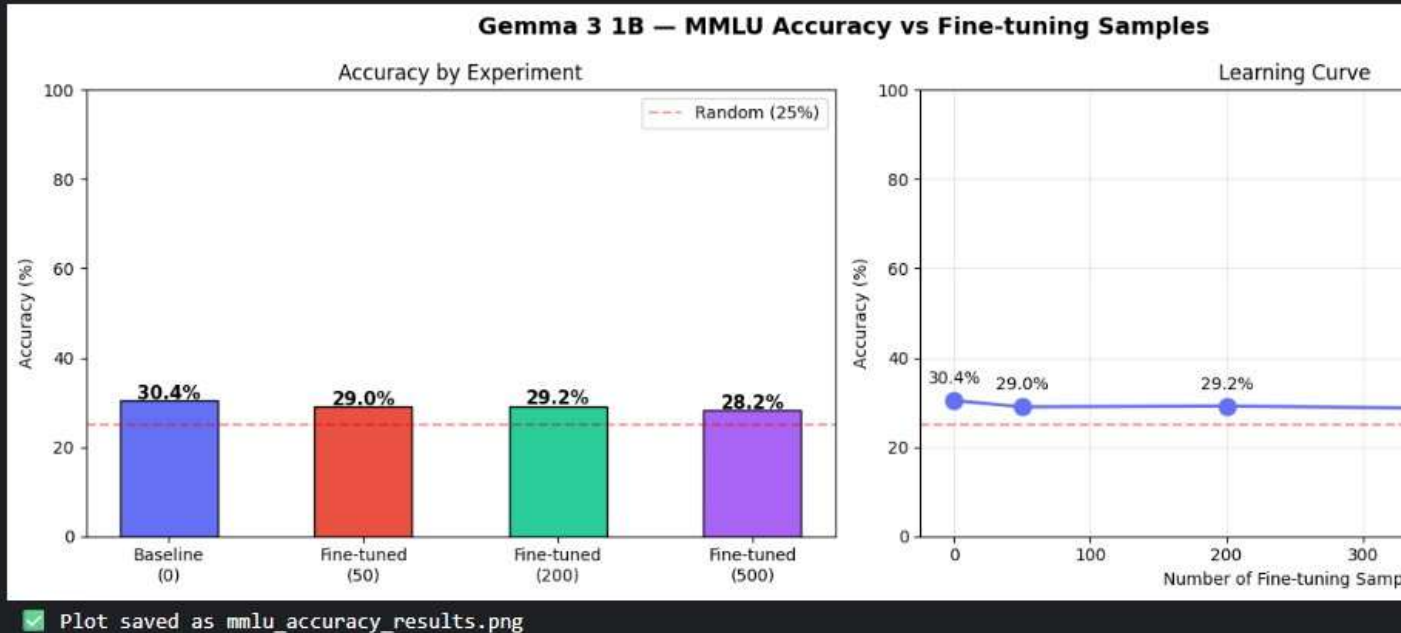


Figure 1: Gemma 3 1B — MMLU Accuracy vs Fine-tuning Samples

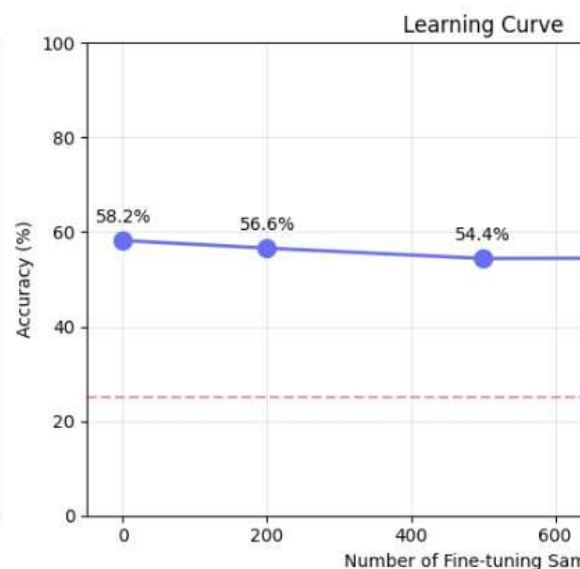
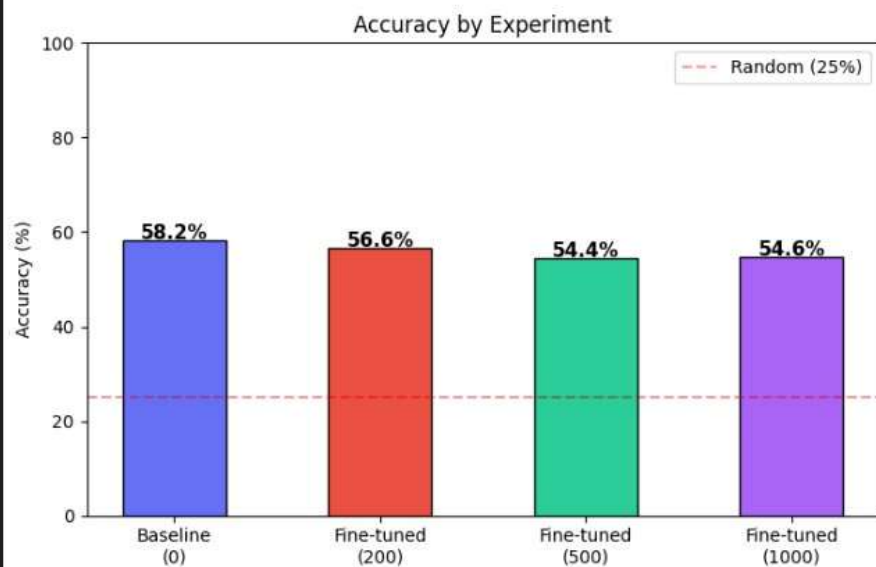
3.2 Gemma 3 1B on HumanEval (Coding)

Experiment	Samples	Accuracy (%)
Baseline (no fine-tuning)	0	9.15%
Fine-tuned	50	7.93%
Fine-tuned	200	9.76%
Fine-tuned	500	8.54%

FINAL RESULTS SUMMARY

Experiment	Accuracy
Baseline (0 samples)	58.20%
Fine-tuned (200 samples)	56.60%
Fine-tuned (500 samples)	54.40%
Fine-tuned (1000 samples)	54.60%

Gemma 3 1B — MMLU Accuracy vs Fine-tuning Samples



✓ Plot saved as mmlu_accuracy_results.png

Figure 2: Gemma 3 1B — HumanEval Accuracy vs Fine-tuning Samples

3.3 Phi-4 Mini on MMLU

Experiment	Samples	Accuracy (%)
Baseline (no fine-tuning)	0	58.20%
Fine-tuned	200	56.60%
Fine-tuned	500	54.40%
Fine-tuned	1000	54.60%

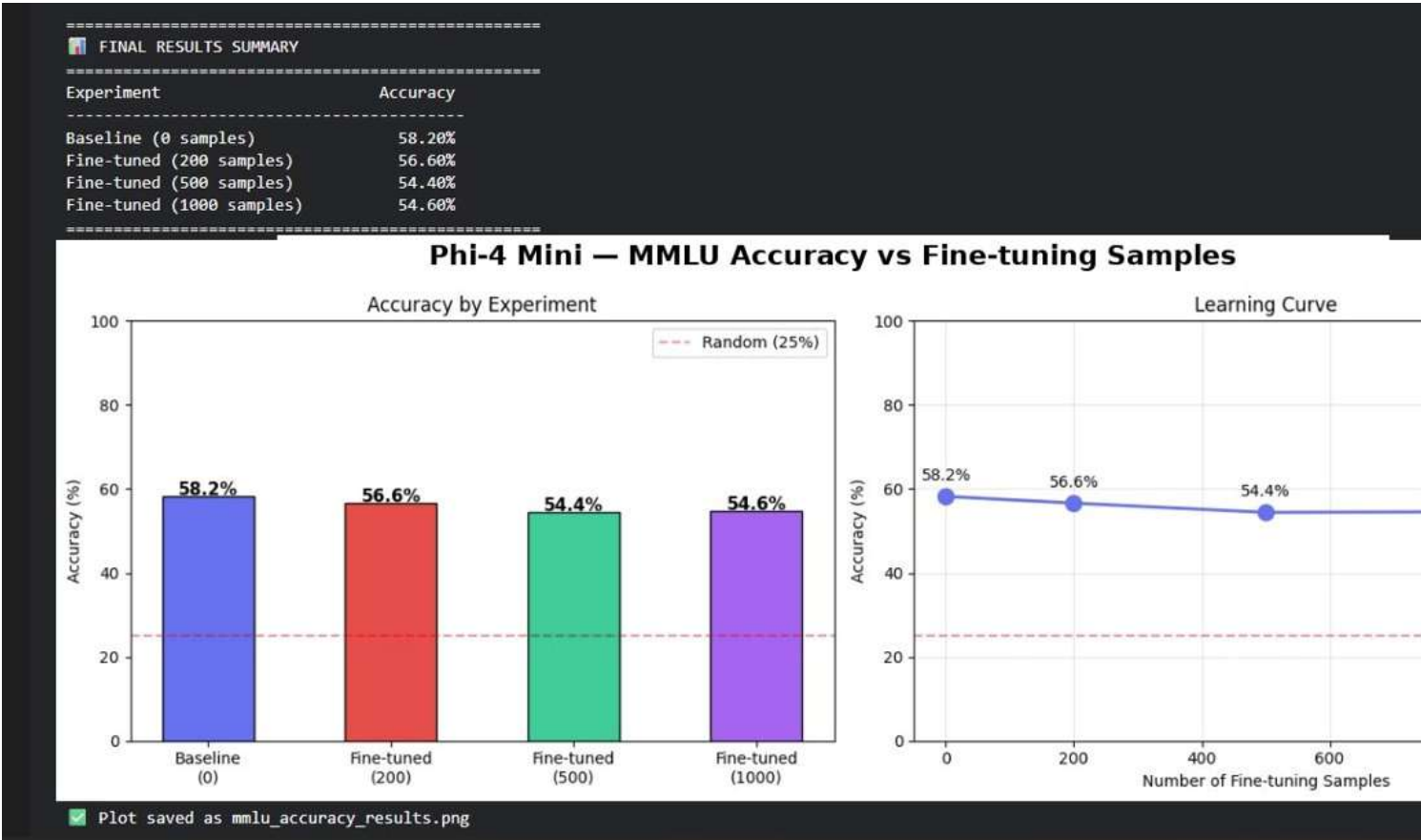


Figure 3: Phi-4 Mini — MMLU Accuracy vs Fine-tuning Samples

4. Analysis & Key Findings

Finding 1: Fine-tuning with small data hurts both models

Across all three experiments, accuracy dropped slightly after fine-tuning compared to the baseline. This is a known phenomenon called catastrophic forgetting — when a model is trained on a narrow subset of data, it partially overwrites the broad general knowledge it learned during pre-training.

Finding 2: Stronger models start higher but fall further

Phi-4 Mini started at 58.2% on MMLU (well above random chance of 25%), while Gemma 3 1B only reached 30.4%. However, Phi-4 Mini also showed a steeper relative drop after fine-tuning, suggesting larger pre-trained models have more general knowledge to lose.

Finding 3: More data does not always help

For Gemma 3 1B on HumanEval, fine-tuning with 200 samples (9.76%) slightly recovered from the 50-sample dip (7.93%), but never surpassed the baseline (9.15%). On coding tasks, the model needs domain-specific training data that closely matches the evaluation format.

Finding 4: QLoRA is effective for memory-constrained GPUs

All experiments ran successfully on a 16GB P100 GPU using 4-bit quantization. Training 500 samples took approximately 20-30 minutes, making this approach practical for anyone with free Kaggle GPU access.

5. Conclusion

This study demonstrates that fine-tuning pre-trained language models with small datasets (50-1000 samples) using QLoRA does not reliably improve benchmark accuracy — and in many cases slightly reduces it. The models already encode strong general knowledge from pre-training that is hard to improve without large, well-curated, task-specific datasets.

The practical takeaway is that for benchmark tasks like MMLU and HumanEval, the base model's pre-training quality matters far more than small-scale fine-tuning. Future work should explore larger fine-tuning datasets (10K+ samples), instruction tuning formats, and task-specific data curation to achieve meaningful accuracy improvements.
